Lalith Uriti

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Education

University of North Carolina at Charlotte December 2026

Bachelor of Science in Computer Science, Systems & Networks Concentration

- Systems Fundamentals, Data Structures & Algorithms, Computer Architecture, Networks, Databases, Object-Oriented Programming, Machine Learning, NLP, Computer Vision, Discrete Math
- Association for Computing Machinery (ACM), CodePath, Charlotte Machine Learning Lab, UNCC SIGGRAPH

Experience

Software Engineering Intern May 2026 – August 2026

CODiT Technologies | Detroit, MI

- Benchmarked LLM answer accuracy on **120+ queries** by building a **RAG pipeline** for document chunking, **embedding-based vector retrieval**, and prompt grounding.
- Cut manual workflow execution time by **35%** by implementing **agent-based workflow orchestration** in Python to coordinate API calls, data retrieval, and multi-step tasks.
- Reduced **GenAI** application latency by **27%** by optimizing **backend API integrations**, retrieval logic, and model requests while checking outputs against predefined quality metrics.

Machine Learning Research Assistant Jan 2026 – Present

Charlotte Machine Learning Lab | With Dr. Razvan Bunescu | Charlotte, NC

- Refined the schema of a **heterogeneous knowledge graph** with **943 nodes and 2,524 edges** by defining typed relationships among knowledge components, assessment questions, and answer nodes.
- Built a **PyTorch GNN pipeline** for **two knowledge-tracing targets**, concept mastery and misconceptions, using synthetic learning sequences from **xKT-Sim**.
- Cut trace-navigation overhead by **45%** by automating simulator trace loading and building interactive **Altair** views, then carried the trace schema and visualization workflow into **CourseLens**.

Undergraduate Teaching Assistant August 2025 – December 2026

College of Computing & Informatics | Charlotte, NC

- TA'd Data Structures & Algorithms; pathfinding, dynamic programming, graphs, trees, hashing, recursion, etc.
- TA'd Computer Networks; TCP/IP, UDP, DNS, routing, switching, firewalls, network security, etc.
- Led labs, reviewed and debugged student code, graded assignments, and supported course logistics and instruction

Projects

Lorca | *Rust, SQLite, Ollama, llama.cpp, Context Managment, Persistent Memory* Architecture

- Built a **model-agnostic AI agent harness** in Rust, emphasizing model intelligence from execution, permissions, and verification through a unified architecture, validated with a native comprehensive automated test suite.
- Designed and implemented **model-driven agent loop** with dynamic tool discovery, file and shell tools, structured schema injection, workspace isolation, and rigird approvals with execution verification.
- Designed **intelligent context management** with adaptive token budgeting, checkpoint-based compaction and persistent **SQLite FTS5** memory with indexed search and paginated recall.
- Integrated **Ollama and llama.cpp** through a unified inference interface for local model experimentation with hardware-aware memory estimates and ongoing research into **weight-space model editing** for self-improvement.

CourseLens | *React, TypeScript, Zod, Apache EChart, Neural Nets, Synthetic Data* Demo

- Built a **teacher-centered learning analytics platform**, transforming student activity and learning mastery into real-time visualizations, misconception tracking, and personalized feedback.
- Developing **student-specific neural knowledge tracing models** and efficient **long-term learning memory**, enabling targeted data retrieval, individualized mastery predictions, and adaptive instruction across each student's learning history.

Blog | *Jekyll, Frontier AI*

- Documenting work, exploring technical concepts, and sharing insights on frontier AI and emerging technologies.
- Local LLMs, context and memory management, model editing, weight-space programming, multimodal learning, inference optimization, harnesse architecture, open-source, etc.

Technical Skills

Languages: Python, Rust, Swift, SQL, Java, C++
Cloud & Data: AWS (EC2, S3), FastAPI, PostgreSQL, PySpark, Pandas, NumPy
AI & ML: RAG, LLM evaluation, Ollama, llama.cpp, PyTorch, Graph Neural Networks
Systems: Linux, Git, Docker, Kubernetes